

Curriculum Vitae

Alon Gurman

CONTACT

alongurman@gmail.com
+1-347-272-4464
[alongurman.github.io](https://github.com/alongurman)
ORCID: 0009-0004-2434-8682

EMPLOYMENT

Postdoctoral Researcher (Joint Rutgers-Flatiron Postdoctoral Fellow)
Department of Physics and Astronomy, Rutgers University, New Jersey, 2026-present

EDUCATION

Ph.D. in Physics and Astronomy
Tel Aviv University, Tel Aviv, Israel, 2020-2026

B.Sc. in Mathematics and Physics
Tel Aviv University, Tel Aviv, Israel, Magna Cum Laude, 2016-2020

Visiting Ph.D. Student
Center for Computational Astrophysics, Flatiron Institute, 2024-present

PUBLICATIONS

Gurman, A., Hu, C. Y., Grudic, M. Y., Steinwandel, U. P., Sternberg, A. “The GHOSDT Simulations: II. Missing Molecules in Simulations of a Self-Regulated Interstellar Medium”. *ApJ*, 995, 186 (2025).

Gurman, A., Sternberg, A., Bialy, A., Cochrane, R. K., Stern, J. ”Molecular Hydrogen in High-redshift Damped Lyman- α Absorbers”. *ApJ*, 995, 116 (2025).

Gurman, A., Steinwandel, U. P., Hu, C. Y., Sternberg, A. “The GHOSDT Simulations: I. Magnetic Support in Gas Rich Disks”. *ApJ*, 984, 142 (2025).

Gurman, A., Hu, C. Y., Sternberg, A., van Dishoeck, E. F., “[CII] Emission in a Self-Regulated Interstellar Medium”. *ApJ*, 965, 179 (2024).

Sternberg, A., Bialy, S., **Gurman, A.**, “HI in Molecular Clouds: Irradiation by FUV Plus Cosmic Rays”. *ApJ*, 960, 8 (2024).

Sternberg, A., **Gurman, A.**, Bialy, S., “HI-to-H2 Transitions in Dust-free Interstellar Gas”. *ApJ*, 920, 83 (2021).

ACADEMIC HONORS

Tel Aviv University scholarship for excellence in teaching, 2024

Tel Aviv University scholarship for excellence in teaching and research, 2022

Tel Aviv University scholarship for outstanding academic performance in graduate studies, 2021

SEMINAR TALKS

Journal Club Talk, University of California San Diego, San Diego, CA, 2025
Institute for Theory and Computation Luncheon, Harvard-Smithsonian Center for Astrophysics, Cambridge, MA, 2025

NY Area Galaxy Seminar, Center for Computational Astrophysics, New York, 2025
 University of Illinois Urbana-Champaign, Illinois, 2025
 CCA Coffee Talks, Flatiron Institute, New York, 2025
 Stars and Interstellar Medium group seminar, Center for Computational Astrophysics, Flatiron Institute, New York, NY, 2024
 Star Formation/ISM Rendezvous seminar, Princeton University, Princeton, NJ, 2024
 Astrophysics Seminar, Tel Aviv University, Tel Aviv, Israel, 2024
 Astrophysics Seminar, Technion – Israel Institute of Technology, Haifa, Israel, 2024
 Galaxy Formation Seminar, Center for Computational Astrophysics, New York, 2023

**CONFERENCE
TALKS**

Geometry of Flows Meeting, New York University, New York, 2026
 The Magnetized Turbulent Universe, Playa del Carmen, Mexico, 2025
 Far-Infrared Fine-Structure Line Workshop, Winona, Minnesota, 2025
 VICO-CICO-CCA Cosmic Origins Workshop, Flatiron Institute, New York, 2024
 ASCOS - Astronomy and Cosmology Conference, Weizmann Institute of Science, Rehovot, Israel, 2024

**TEACHING
AND
MENTORING**

Teaching Assistant, Introduction to Astrophysics (physics final year), 2020-2025
 Teaching Assistant, 2nd year physics lab, 2020-2025
 Co-advisor, Matan Berkovich, Technion – Israel Institute of Technology, Haifa, Israel, 2025-present
 Professional Development for Teachers and Development of Teaching Materials, joint program by Tel Aviv University, Israeli Space Agency, and the Israeli Ministry of Education, 2021-2023
 Mentor, Meduyakot program for women in exact sciences, Tel Aviv University, 2021

**PROFESSIONAL
EXPERIENCE**

Reviewer in Training, Monthly Notices of the Royal Astronomical Society, 2023-present
 Research Assistant, MRI motion correction with Frydman Group, Weizmann Institute of Science, 2018-2019

**CLIMBING
AND MOUNTAINEERING**

Traveled across the Middle East, Europe, and USA, climbing big walls at elite level such as El Capitan in Yosemite National Park CA, Mt. Hooker in the Wind River Range WY, and Djebel Nasrani in Wadi Rum, Jordan
 U19 sport climbing Israeli national champion, 2011